

CONTROL UP TO 16 APPLIANCES Without Using A Microcontroller

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Generally, the total number of appliances that can be controlled without using a microcontroller (MCU) is limited to three or four. One possible solution to this limitation is described in this project. Using this circuit, you can control up to 16 electrical appliances using HT12E and HT12D in transmitter and receiver sections, respectively. The ICs convert BCD parallel data into serial data. Since HT12E can take BCD inputs, you can feed ($2^4 = 16$ different types) many inputs that can be decoded at the receiver section easily.

Circuit and working

The circuit is divided into three main sections: transmitter, receiver and bistable.

Transmitter section. The circuit diagram of the transmitter section is shown in Fig. 1. It is built around 4x4 matrix keypad connected across connector

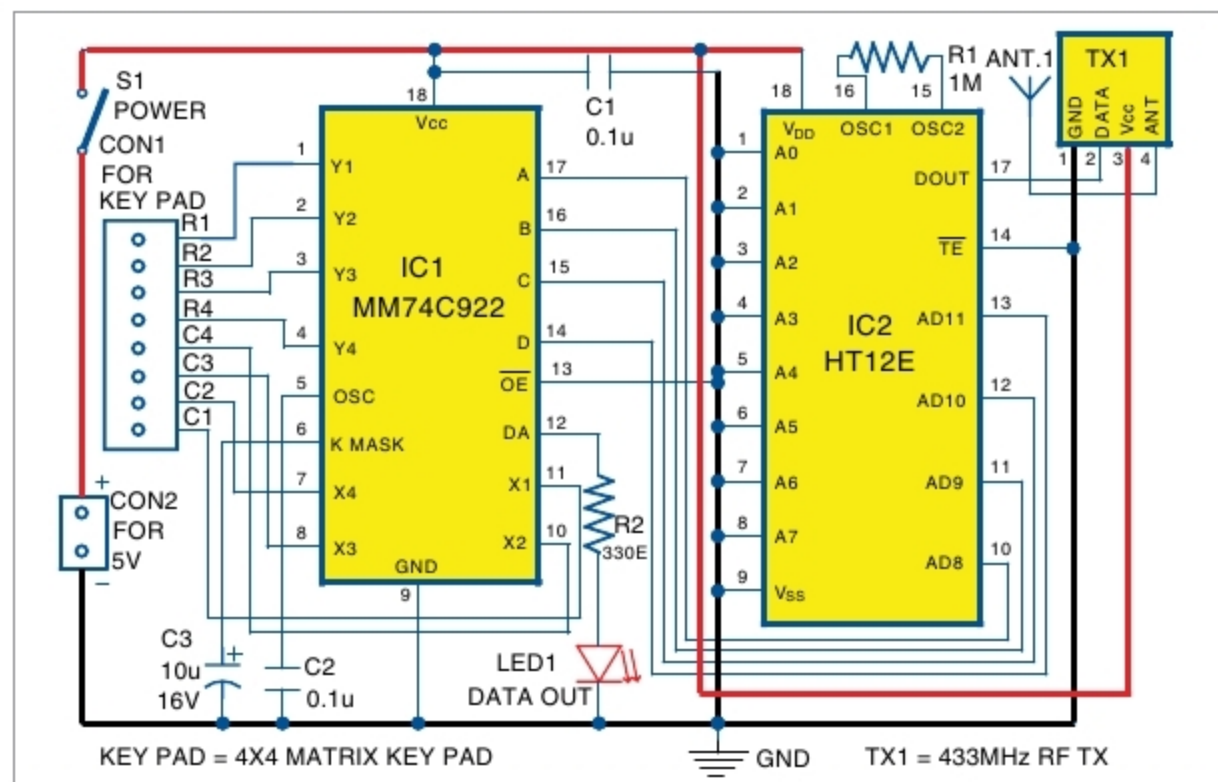


Fig. 1: Circuit diagram of transmitter section

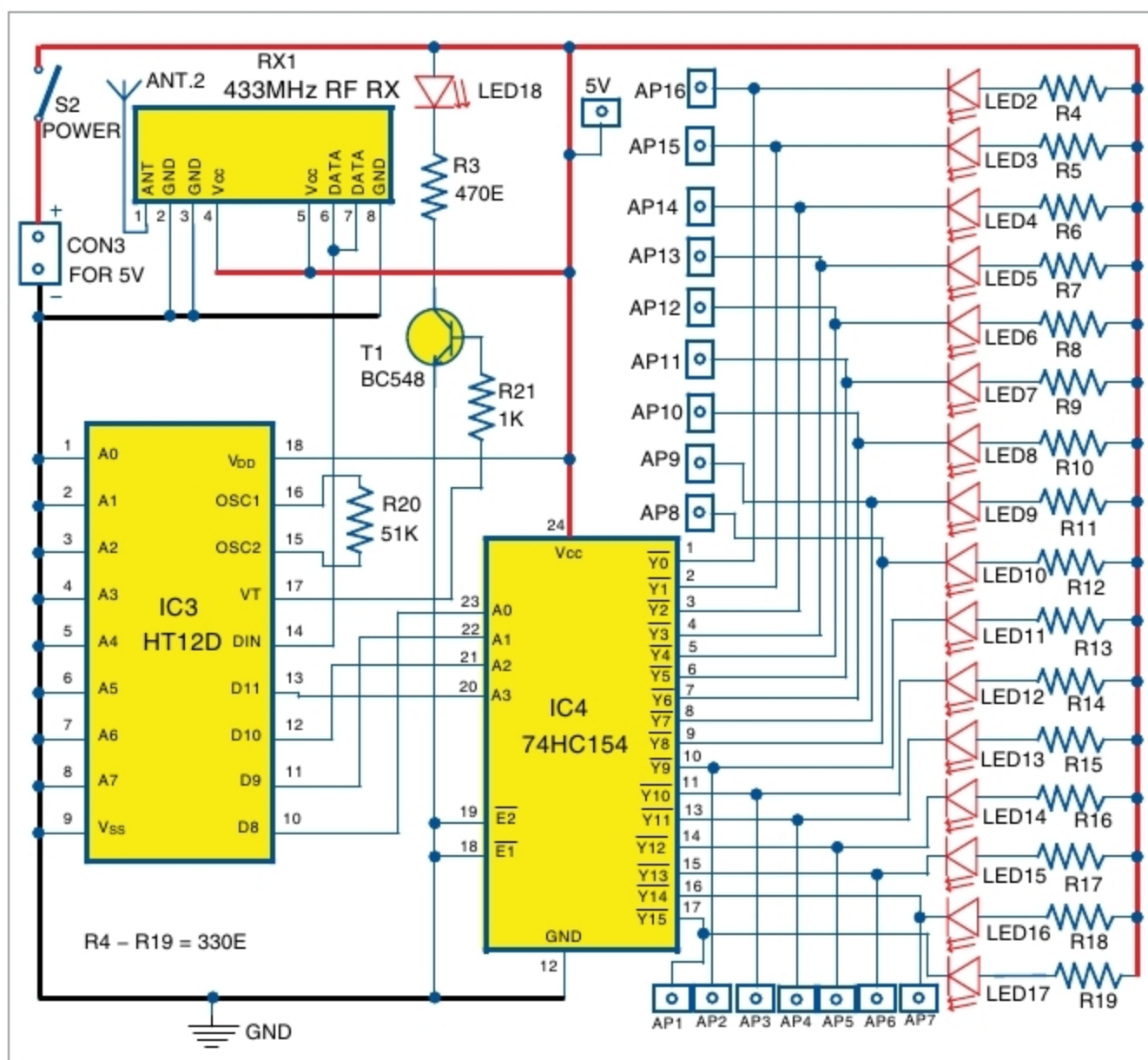


Fig. 2: Circuit diagram of receiver section

CON1, key encoder MM74C922 (IC1), encoder HT12E (IC2), 433MHz RF transmitter module (TX1) and a few other components. Here, MM74C922 is used to get BCD output on pressing any button of the keypad.

Receiver section. The circuit diagram of the receiver section is shown in Fig. 2. It is built around 433MHz RF receiver module (RX1), decoder HT12D (IC3), 4- to 16-line decoder 74HC154 (IC4) and a few other components. Here, 74HC154 is used to decode each BCD code into corresponding output (AP1 through AP16).

Signal received at the receiver side is active low till a pushbutton (input from keyboard) is pressed at the transmitter section. In order to control an appliance with this signal, make a